Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/10/2007
PI Name:	Immer, Christopher Ph.D.		
Project Title:	Advanced Fire Detection In Enclosed Environment Via The Detection Of Chemical Vapor Signatures From Pre-Combustion Processes Using Machine Olfaction		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	none available	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	
Organization Name:	ASRC Aerospace Corp.		
PI Address 1:	Mailstop ASRC-20		
PI Address 2:			
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City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899	**Congressional District:**	24
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-B
Start Date:	10/01/2005	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Greenberg, Paul	**Contact Phone:**	216-433-3621
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	PI changed to Christopher Immer from Bruce Linnell, sometime during FY2006, per GRC contact information. See LINNELL for previous reports (12/07).		
COI Name (Institution):			
Grant/Contract No.:	NNC04CB08C		
Performance Goal No.:			
Performance Goal Text:			
	The detection of fires is very important within closed environments such as the Space Shuttle and International Space Station. Fires need to be detected within such areas as soon as possible, preferably before combustion has taken place. Previous research performed at KSC has shown that the multi-sensor technology commonly known as the electronic nose (e-nose) is able to detect the offgassing products from the insulation of heated wires well before combustion starts. Note that e-nose technology is not concerned with identifying in an analytical sense the chemicals being detected, but with classifying a mixture of vapors as coming from one source or another. This project would expand upon these results, and produce an integrated system of sensors and software which would not only provide a pre-combustion alarm, but also identify what kind of material is being overheated (wire insulation, printed circuit board, etc.). Metal oxide semiconductor (MOS) sensors are low cost, available in numerous miniature configurations, have a wide range of		

Task Description:	selectivity to various vapors, and are simple to operate. To that end, commercially available MOS sensors will be incorporated into a sensor array and exposed to thermal out-gassing products of electrical components. In addition, conventional circuit board components (e.g., IC chips-usually encapsulated in resins) will be subjected to heat cycles to mimic catastrophic failure with a potential fire hazard. Pattern recognition techniques will be developed at KSC to identify those commercially available sensors which provide the most sensitivity and reliability to detection and identification. It is recognized that other sensor technologies (e.g., SAW, quartz micro-balance, etc.) could potentially increase the identification power of the array, and so other sensor types will be tested and incorporated into the array if they are found to be useful. While this technology would be designed with flight consideration in mind, the result of this project will not be a flight-ready system, but one or two prototypes whose sensors have been partially screened for flight qualification.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Fire detection systems.
Task Progress:	The electronic nose has been adopted for use as a pre-combustion fire detection system that would specifically look for faults electronic components and electrical wiring. Specifically, the Airsense Enose has been selected and 4 classifiers have been trained to look at 9 different electrical components and wiring. For components in a 1'x1'x1' volume, the Enose/Classifers can detect the presence of pre-combustion gases within 2 minutes. Additionally, it is most likely possible to identify the source of the gases (e.g. heated wire, resistor, inductor, etc…). A prototype portable unit has been developed that implements the fire detection mechanisms. This unit is self-contained, operates on standard 120V 60Hz AC, and can audibly, visually, and electrically alarm upon the detection of pre-combustion components. Varying background gas and pre-combustion gas concentration would likely limit the operation of this technology to very specific, well controlled environments. However, given those more specific, well controlled conditions, the Enose would successfully work as a pre-combustion fire detection system.
Bibliography Type:	Description: (Last Updated:)